HEAPLINE · REGULATORY AND STANDARDS REGISTER

HeapLine regulatory and standards register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	First register, written for the kit 1.9.1 upgrade; standards and guidance checked on their pages on 2026-11-03

HeapLine regulatory and standards register

HeapLine is a community-held kit of probes, boards, a stretcher and a hand capstan for the first minutes after a dumpsite slide; no regulation was found that certifies it as a whole, and it is published as an open engineering reference, not certified rescue equipment. The design leans on lifting-accessory, machinery-safety and protective-equipment standards for its rope, sling, shackle and glove choices. This is not legal advice; anything marked "to assess" needs a qualified reviewer before any product claim. Method: STANDARDS section 23.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	ISO 12100:2010 Safety of machinery, general principles for design, risk assessment and risk reduction	International	Method for hazard identification, risk estimation and risk reduction	The hazard analysis and design FMEA in HPL-RSK-001 follow its order: remove by design, guard, then inform	Design lead	Applies; used as method (HPL-RSK-001)
G2	ISO 11228-2:2007 Ergonomics, manual handling, Part 2: pushing and pulling	International	Recommended limits for whole-body pushing and pulling	Bar push of 114 N each (PR-004), stretcher haul of 127 to 189 N each (PR-005) and, for options B and C, the rope and lever forces in HPL-STM-001	Design lead	To assess: tables not read; check at TRL 4
G3	EN 1492-2:2000+A1:2008 Textile slings, safety, roundslings made of man-made fibres	EU, widely used elsewhere	Rating and testing of polyester roundslings; excludes lifting people	The 2 t round slings (BOM line 21) anchor the capstan to a tree, vehicle or column and choke debris: a pulling anchor, never a lift, never a person	Design lead	Applies to the bought sling; use as a pulling anchor to assess
G4	EN 13889:2003+A1:2008 Forged steel shackles for general lifting purposes, dee and bow shackles, grade 6	EU, widely used elsewhere	Dimensions and working load limits of grade 6 shackles with threaded pins	Bow shackles WLL 2 t with 19 mm pins on the sling and the capstan eye (BOM line 21); the model shows them with a plain pin	Design lead	To assess: buy rated shackles with threaded pins at TRL 4
G5	EN 388:2016+A1:2018 Protective gloves against mechanical risks	EU, widely used elsewhere	Abrasion, cut, tear and puncture classes for gloves	PPE line 26 asks for cut-resistant gloves at level C or better; the cooperative supplies them from its own stock	Cooperative (PPE)	Applies to the bought gloves; levels to assess against the waste on site
G6	ISO 4017, ISO 4032 and ISO 7089 hex bolts, nuts and washers, as named by the CAD parts library	International	Dimensions of the bolts, nuts and washers in the model	The 51 fasteners in the 347-part model of option A (`cad/fidelity.json`); numbers as named in `.kit/cad_parts.py`, not checked against the standards this session	Design lead	Applies; to check at TRL 4
G7	Landfill gas safety guidance, for example	United States (guidance),	Methane explosive range 5	The no-flame, no-smoking rule and the		Applies as guidance;

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	ATSDR Landfill Gas Primer, chapter 3	used worldwide	to 15 % in air; carbon dioxide and other components of landfill gas	stop signal for a gas smell, smoke or hissing (HPL-PRC-001); no detector in the base kit	Co-design partner	local site rules govern
G8	Wound care and tetanus guidance, for example CDC Clinical Guidance for Wound Management to Prevent Tetanus	United States (guidance), used worldwide	Higher-risk wounds (puncture, soil-contaminated) and vaccination status	Sharp waste and soil on a search; the briefing wording and PPE (HPL-STM-001 S20); not medical treatment	Cooperative	Applies as guidance; local health advice governs
G9	Machinery or equipment safety regulation of the country of use	To assess	Whether a hand-powered kit held by a cooperative is "machinery" in law	None identified; no CE-style marking is claimed	Regulatory consultant	To assess
G10	Rescue equipment, stretcher and casualty-handling rules of the country of use	To assess	Whether a made sheet stretcher may be used to move a casualty	None identified; HeapLine is not certified rescue equipment and says so (PR-014)	Co-design partner, with the Red Cross branch or fire service	To assess
G11	Fibre rope and polyester double braid specifications	To assess	Minimum breaking load and testing of the rope	The 14 mm rope is bought on the maker's minimum break load (HPL-STM-001 S8); no rope standard number was checked	Design lead	To assess
G12	Permission to use a tree, vehicle or column as an anchor; rules for working at an active or closed dumpsite	Local	Owner's consent, site access, occupational and public rules for informal waste sites	The sling anchor rule (HPL-DDR-001 item 4); the shed and crate hosted at the cooperative's site	Co-design partner	To assess locally
G13	CERN-OHL-S-2.0 and MIT licences; REUSE	International	Licensing of the hardware and calculators	`REUSE.toml`, `LICENSES/`	Project lead	Met by design

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- ISO, ISO 12100:2010, https://www.iso.org/standard/51528.html, checked 2026-11-03.
- ISO, ISO 11228-2:2007, https://www.iso.org/standard/26521.html, checked 2026-11-03.

- Genorma, EN 1492-2:2000+A1:2008, https://genorma.com/en/standards/en-1492-2-2000-a1-2008/amp, checked 2026-11-03.
- SIS (Swedish Standards Institute), SS-EN 13889+A1:2008, https://online.sis.se/en/produkter/maskinsakerhet-357f77b1/chains-ropes-webbing-slings-and-accessories/ssen13889a12008, checked 2026-11-03.
- Singer, EN 388:2016+A1:2018, https://www.singer.fr/en/norm/en388, checked 2026-11-03.
- ATSDR, Landfill Gas Primer, chapter 3 (2001), https://www.atsdr.cdc.gov/HAC/landfill/html/ch3.html, checked 2026-11-03.
- CDC, Clinical Guidance for Wound Management to Prevent Tetanus, https://www.cdc.gov/tetanus/hcp/clinical-guidance/, checked 2026-11-03.